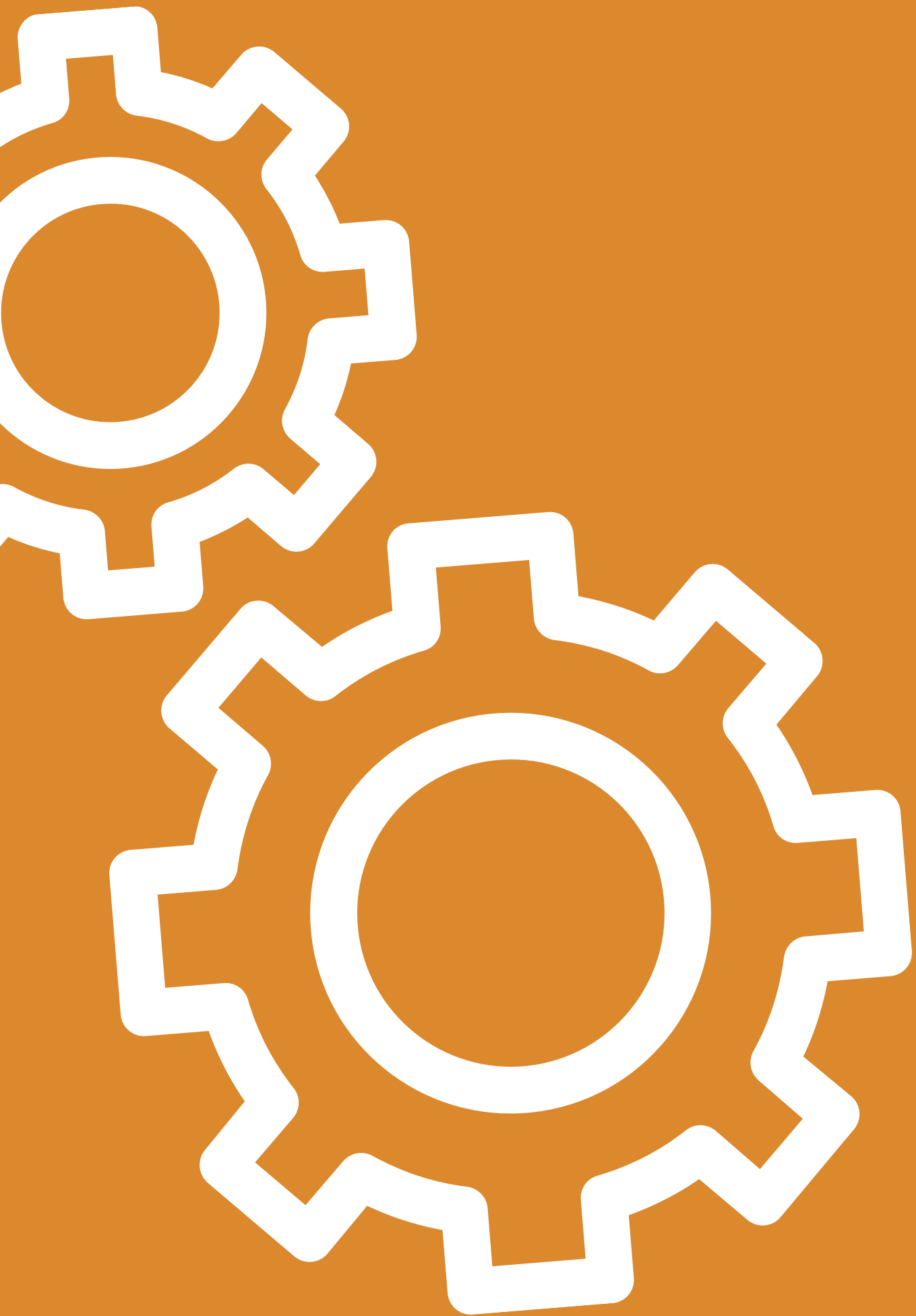




WEIGHT MEASUREMENT



GROUP - 2

Sneha Sivaram
122201003



GROUP LEADER
Circuit Connections,
Calibration, Procuring
components, Content for
PPT

Mausumi Bhuyan
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MECHANICAL LEADER
Deciding dimensions of
weighing scale, Managing
the construction of
mechanical system

Goli Satwika
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Circuit Connections,
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PPT

Konduru Bhanu Chandrika
122201006



**Circuit connections,
Content for PPT**

Sabhavat Sandhya
122201009



**Deciding dimensions of
weighing scale, Managing
the construction of
mechanical system**

Mummidi Yaswant
122201013



**Soldering components,
Calibration, Circuit
Connections, Procuring
components**

Priyanshu Pahi
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**Arduino coding, Calibration,
Circuit Connections**

Amuluru Pavan Sai
122201022



**Designed 3D MECHANICAL
model , Final PPT , Content
for PPT**

Nalan S
122201034



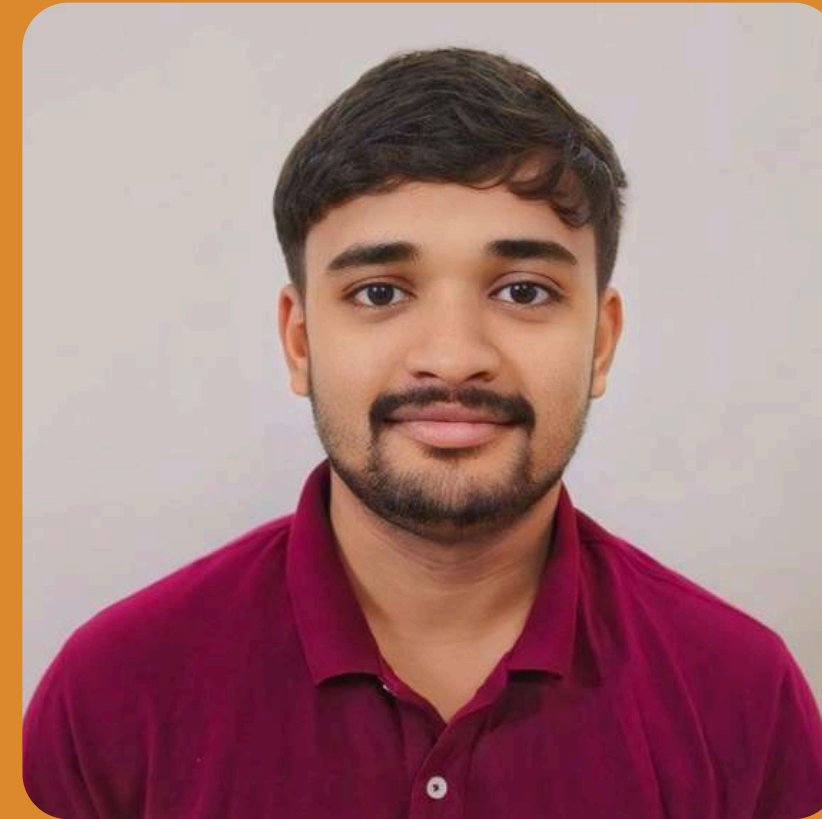
**Manual design and assembly
of components , Material
arrangement**

Sai Charan
122201038



**Manual design , assembly of
mechinal model , Metal
cutting.**

Himanshu Kumar
122201041



**Circuit Connections,
Arduino coding**

DESIGN CONSTRAINTS

& SOLUTIONS



PLATE SHAPE AND DIMENSIONS:

We designed a square plate with upper plate dimensions of 20x20 cm and lower plate dimensions of 30x30 cm. The force bearing end of each load cell is at a corner of the top plate, to ensure better weight measurement irrespective of position of object at the centre or not.

OPTIMAL LOAD CELL POSITIONING:

The four load cells were positioned at 45-degree angles at each corner of the square plate. This ensured that each load cell contributed to measuring the load from different directions, enhancing the overall accuracy of the system.





MATERIAL SELECTION FOR DURABILITY:

An acrylic plate was initially considered, but it could not handle heavier loads without deforming, leading to inaccuracies. The plate material was switched to metal, which provided the required strength and stability for accurate measurements under larger loads.

WEIGHT RANGE:

The weighing machine should measure weights from 0 kg to 18 kg with a resolution of 0.1 kg. Hence, we used four load cells, each rated for 5 kg. Together, they support up to 20 kg when distributed across the platform. Since weight of the top plate is almost 2kg, we can measure only upto 18 kg.



RESOLUTION:

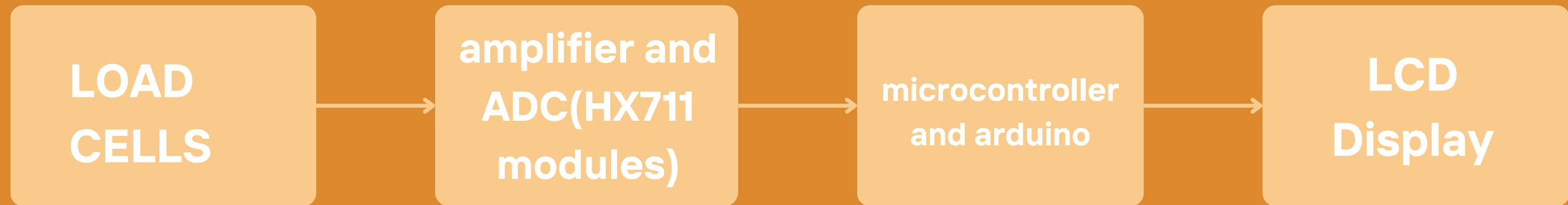
We chose HX711 modules for their 24-bit ADCs, to enable better resolution. We set the HX711 gain of 128 for maximum sensitivity, to capture even small changes in load cell output to achieve 0.1 kg resolution.

We built a circuit to take obtain output from 4 load cells using 4 HX711 modules and also attempted to improve the design by using 1 HX711 module and 2 switches to reduce the hardware.



BLOCK DIAGRAMS

USING 4 HX711 MODULES:



USING 1 HX711 MODULE AND 2 SWITCHES:



DESIGN ▾

SOLID

SURFACE

MESH

SHEET METAL

PLASTIC

UTILITIES

MANAGE

CREATE ▾

AUTOMATE ▾

MODIFY ▾

ASSEMBLE ▾

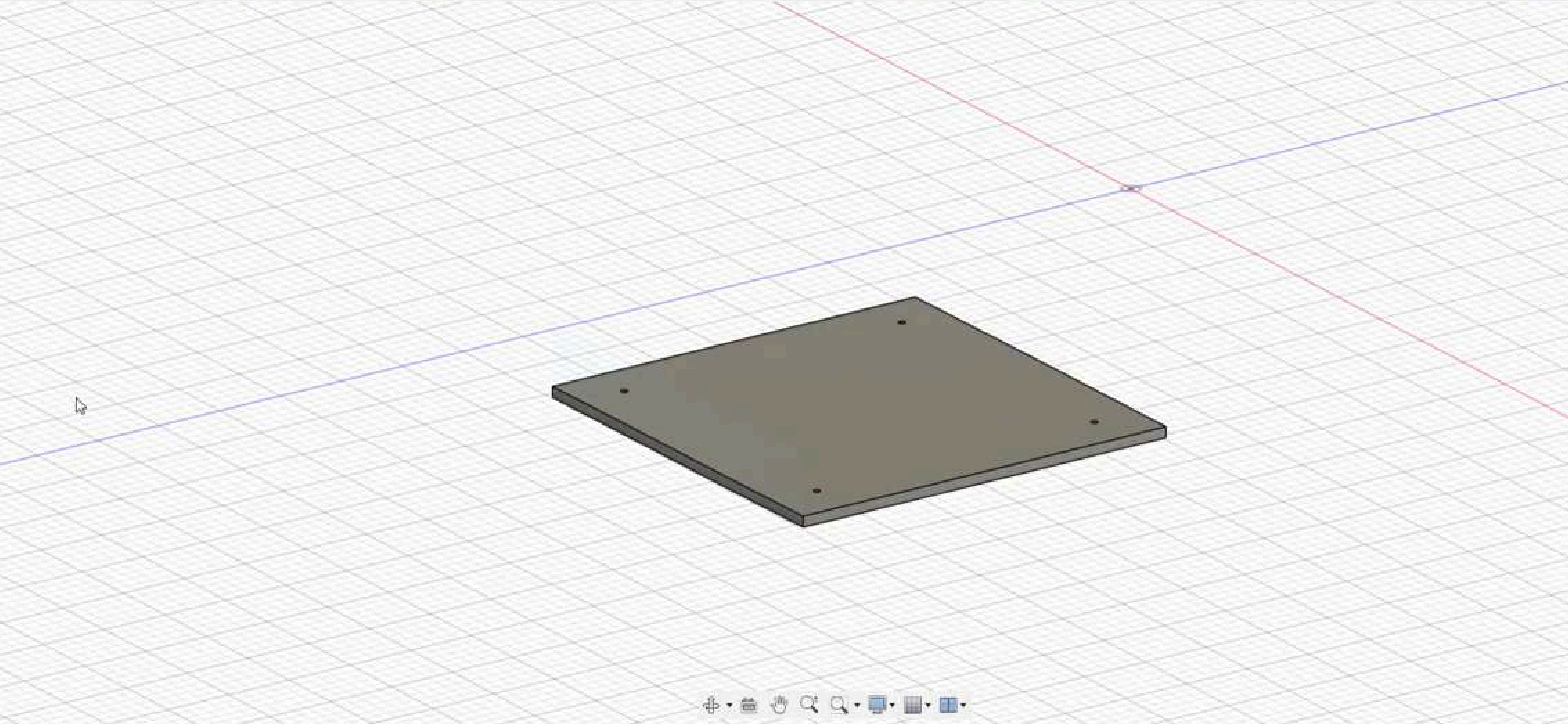
CONFIGURE ▾

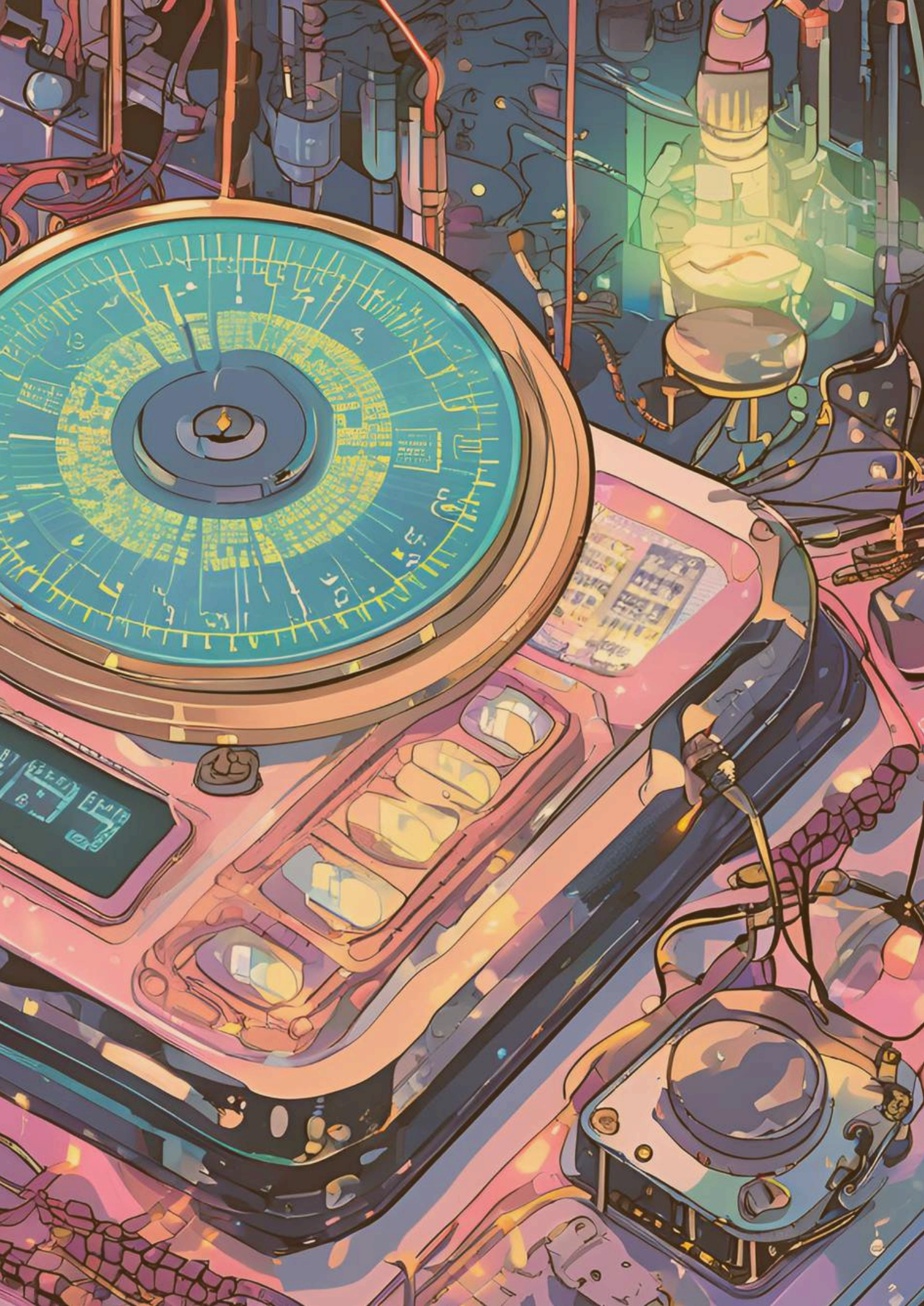
CONSTRUCT ▾

INSPECT ▾

INSERT ▾

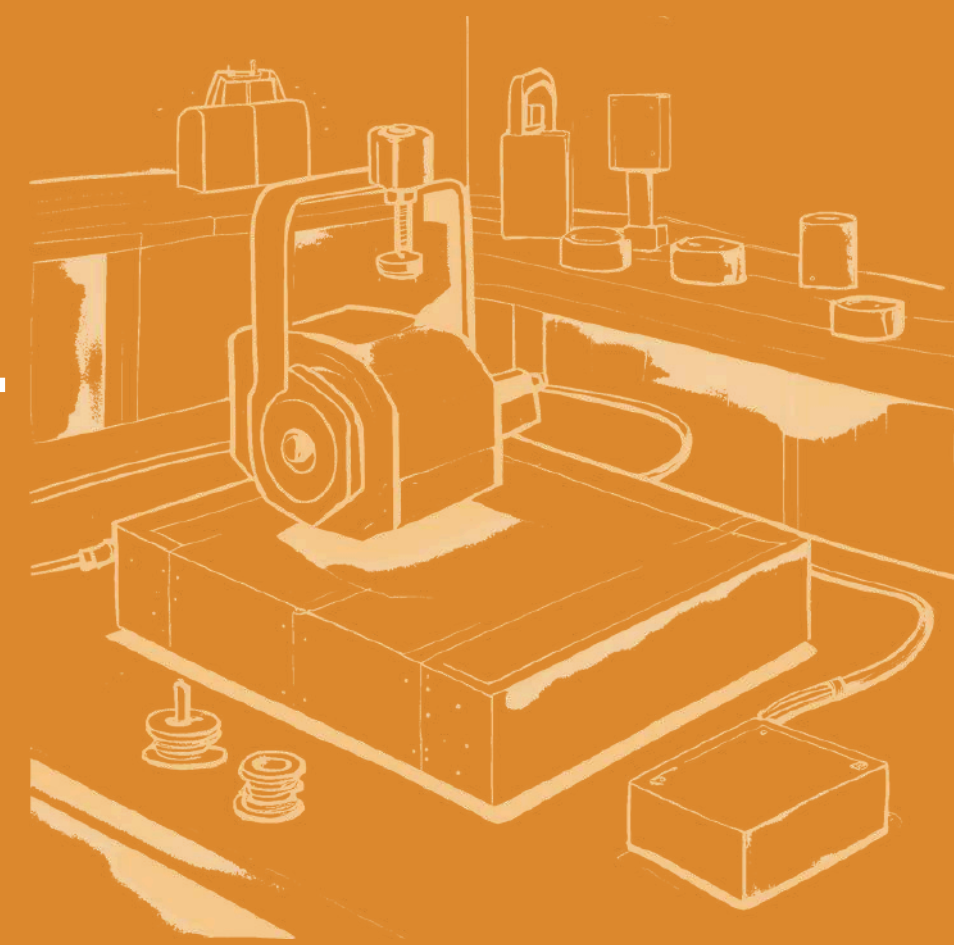
SELECT ▾





EXPERIMENTAL SETUP

ELECTRICAL & MECHANICAL



MECHANICAL

- In our design, the square upper plate has a dimension of 20x20 cm and the lower plate has a dimension of 30x30 cm.
- The four load cells are positioned at 45-degree angles at each corner of the upper square plate to ensure more accurate readings irrespective of position of weight placed.
- Appropriately sized bolts are used as spacers between the load cells and the plates.



ELECTRICAL

CONNECTIONS USING 4 HX711 MODULES

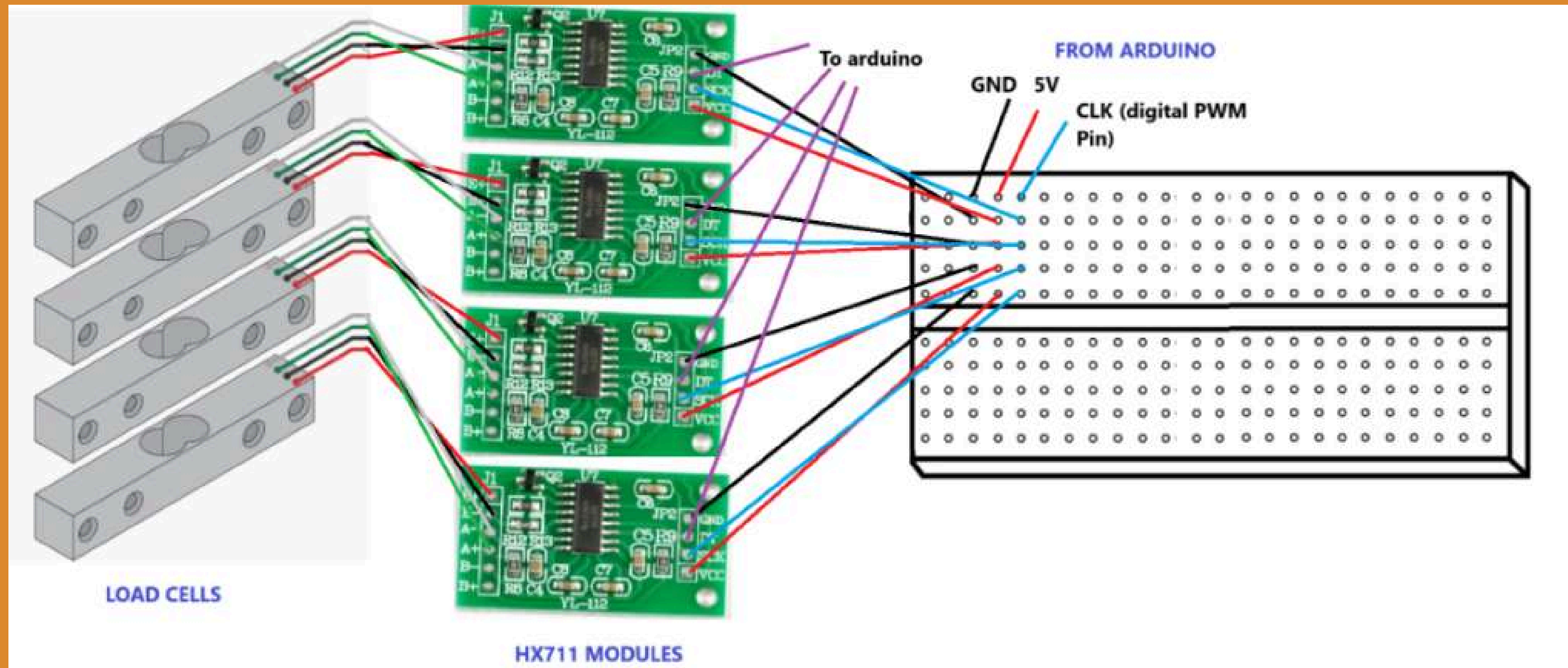
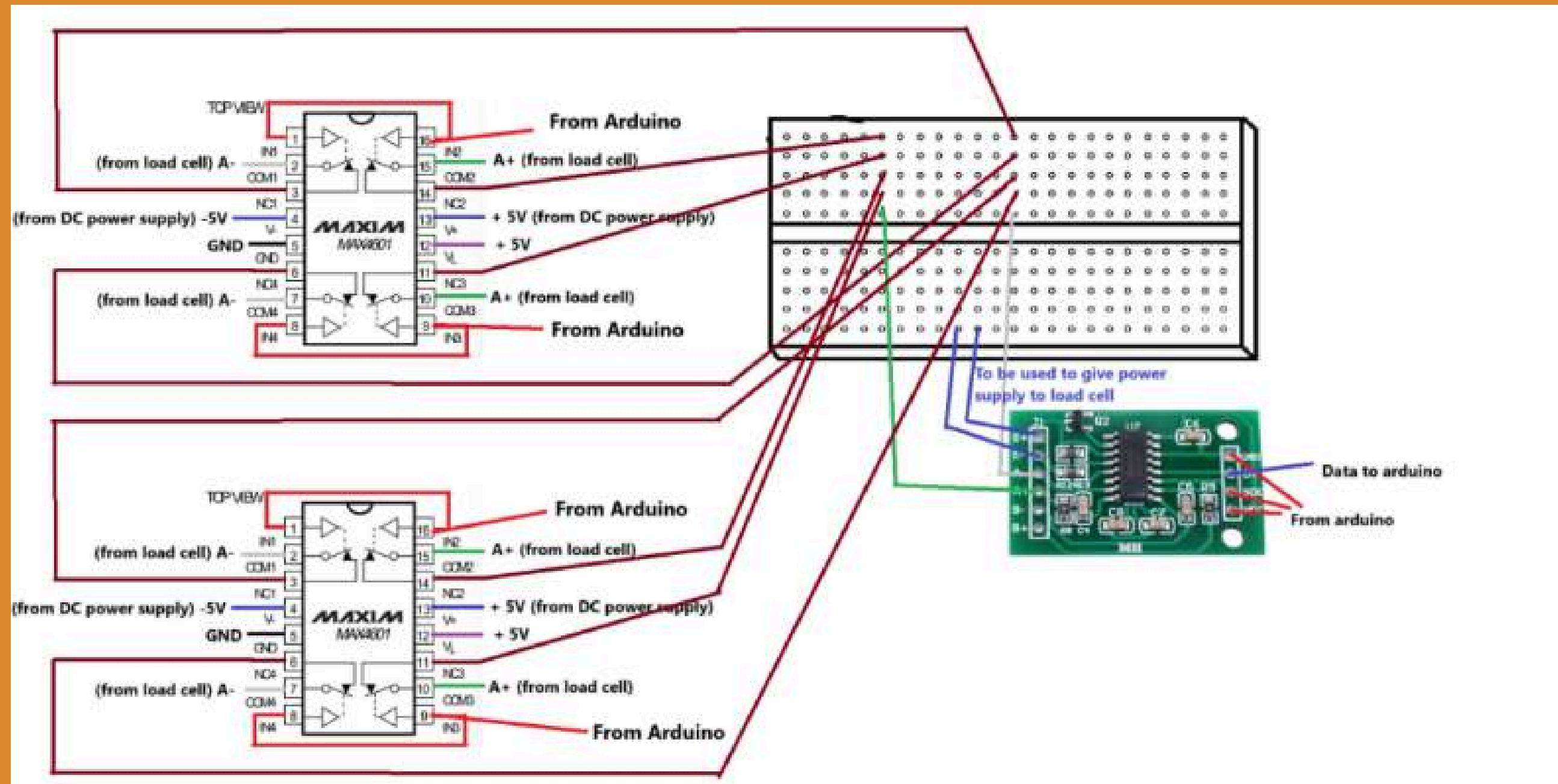


ILLUSTRATION OF SWITCHING CIRCUIT FOR THE CASE OF 4 LOAD CELLS



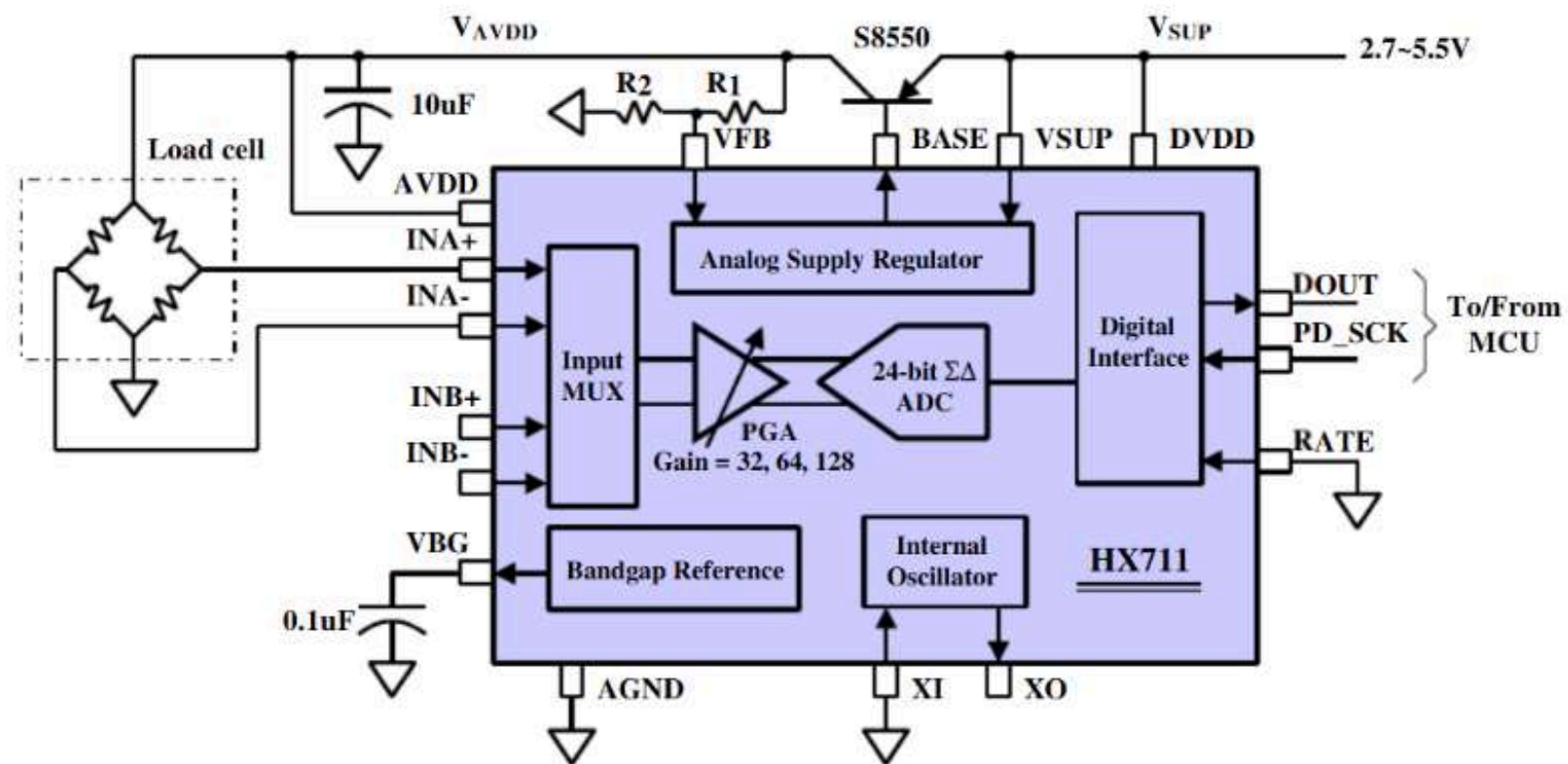


Fig. 1 Typical weigh scale application block diagram

- HX711 IC:
- Has 2 differential input channels with programmable gains.
- Has Low-Noise Programmable Gain Amplifier (PGA) to amplify the small output signals from bridge sensors like load cells (we use a gain of 128).
- Has 24 bit sigma- delta ADC converter
- Has on-chip power supply regulator to provide stable power for the ADC and the connected load cells to ensure minimal noise and reliable operation.
- After the HX711 completes an ADC conversion, it sets the DOUT pin low to indicate that the 24-bit data is ready to be read.
- The microcontroller generates a clock signal on the PD_SCK pin. This is done by toggling the PD_SCK pin between low and high states.
- Each rising edge of PD_SCK shifts out one bit of data through the DOUT pin.

CALCULATION OF CALIBRATION FACTOR



Weight of unknown object = $(r1 - p) * c$

$(p - r) * c$ = Standard weight

c = Standard weight / $(p - r)$

p - Reading with just the plate
r - Reading with standard weight
r1 - Reading with unknown weight



CHALLENGES FACED

Calibration Across Multiple Load Cells:

- We initially calibrated each load cell to give equal contribution to the total weight. While this approach works well when the object is placed at the centre, it doesn't work well away from the centre.
- So we calibrated each load cell individually by placing known weights directly above them at the corners of the top plate.
- In this case, the contribution of each load cell to total weight reading may not be equal but the sum of readings of the individual load cells will give the correct weight and seems to be a better option to get more accurate readings irrespective of the position of object on the load cell.





CHALLENGES FACED

Choice in material of top and bottom plates:

- Initially used acrylic, not strong enough for larger weights. Changed to metal sheets, however the top metal sheet weighs almost 2 kg.
- Due to the top metal sheet being heavy it may be causing a larger deformation in the load cells due to which it cannot be used to measure very small weights.
- Ideally we need to use a lighter metal sheet for the top plate.



QUESTIONS ASKED DURING EARLIER REVIEW

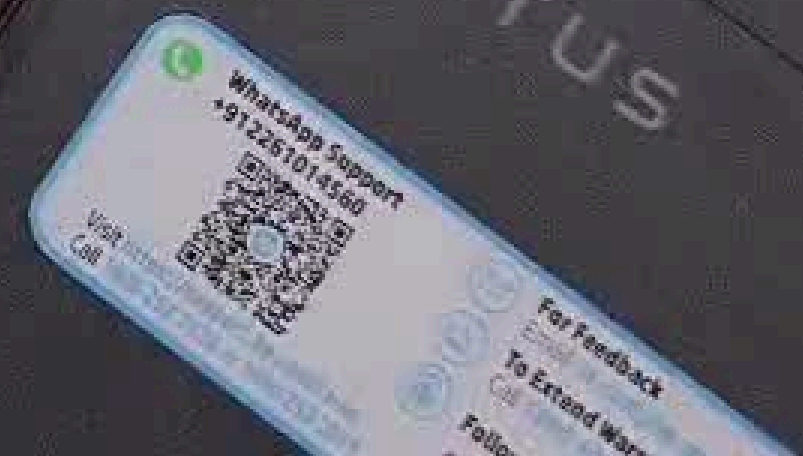
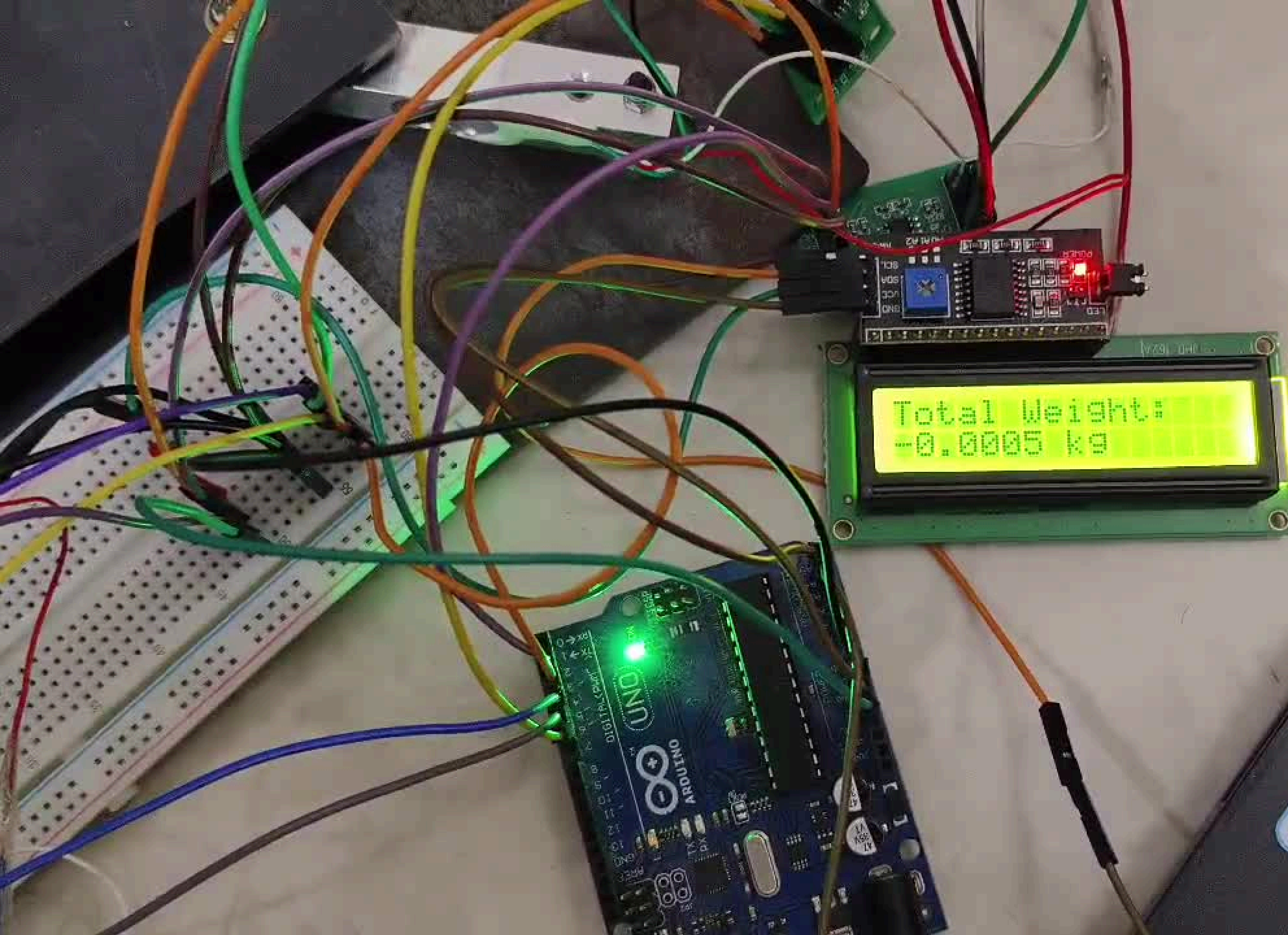
- Addition of flowchart/ diagram to explain electrical design: Block diagram and circuit diagrams have been updated to explain our circuits in a more clear way.
- Use of single HX711 and switches: We tried using one HX711 and two SPST switches but are currently not getting accurate readings from load cells.
- Use of Metal Sheet: Instead of Acrylic, we have made the weighing platform with metal sheets.
- Protocol used for transmission using HX711: Serial communication but does not adhere to standard protocols like I2C or SPI
- Change in calibration: Changed it to calibration of individual load cells which seems to give better results irrespective of location of object on the top plate.

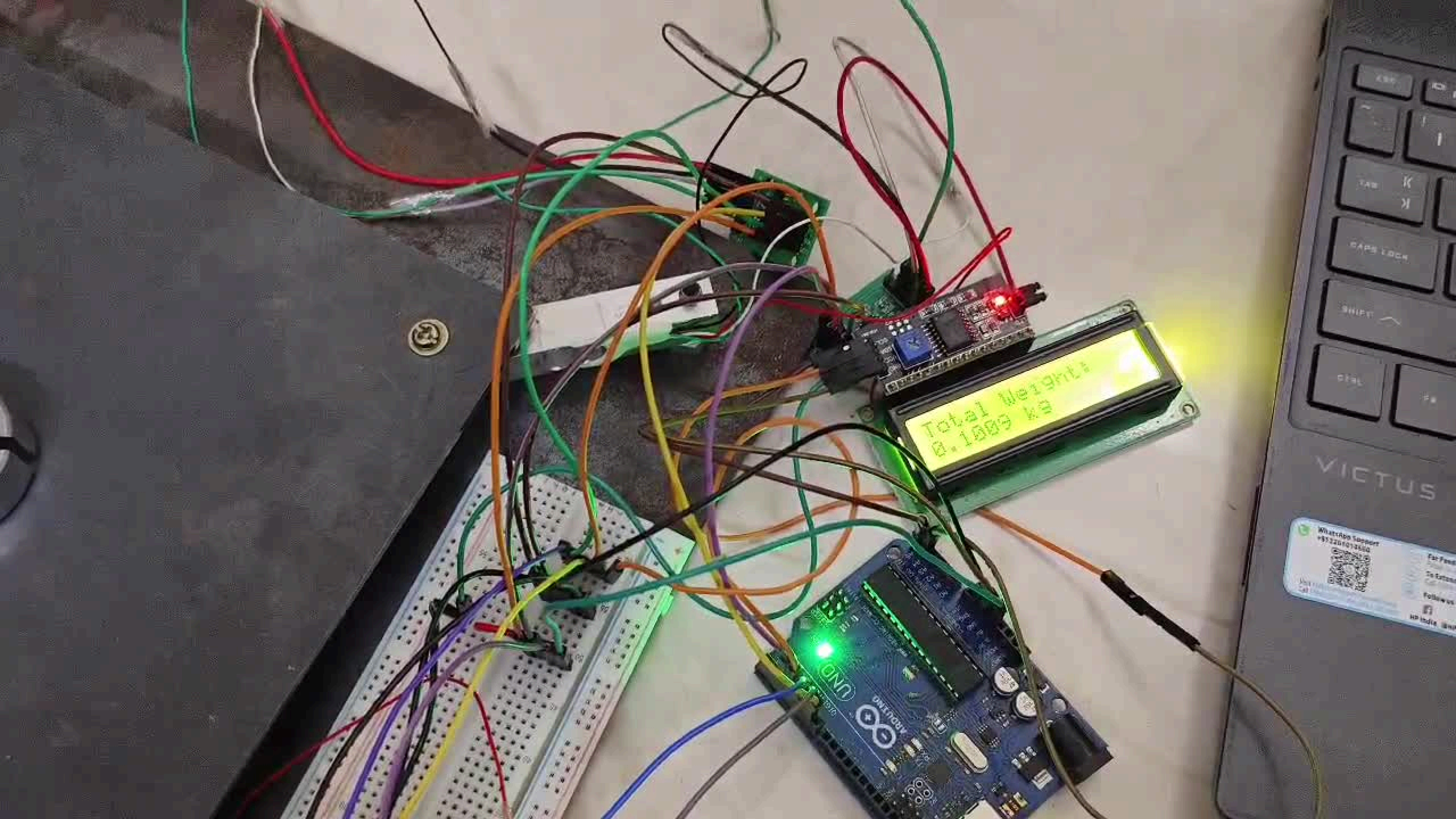
FIELD VALIDATION RESULTS

Range: 0 - 17 kg

Resolution: 50 - 100 g







Total Weight:
0.1000 kg

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UNO

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FUTURE SCOPE



- **Fixing the switching circuit**
- **Addition of filtering**
- **Adding adjustable screw levelers can ensure the weighing machine is stable on various surfaces.**
- **Building a box around the setup**
- **Consolidating electrical components onto a printed circuit board (PCB) reduce wiring complexity, and make the machine more compact.**